

ENAE464 - 0202

Lab01: Pressure Measurements of Inviscid Flow over Cylinder

Due on February 13th, 2026 at 11:59 AM

Dr. Silbaugh, 02:00 PM

Mikołaj Kostrzewa & Vai Srivastava

Experiment Performed: February 6th, 2026

Report Submitted: February 11th, 2026

This is a brief letter accompanying a report indicating that the report is enclosed or attached. In some cases, the letter is put at the beginning of the report following the title page rather than on top of it. It simply is up to your preference whether to make the letter a part of your report by putting it behind the title page or to separate it from the main report by placing it on top of the title page. While it serves not much purpose other than officially turning the report over to the reader, it is a professional and courteous way to present the report.

In general, the letter should be addressed to the person or organization that requested the information contained in your report. For the purpose of this course, however, your letter can be addressed to the laboratory professor whose experiment you just have completed.

Contents

1	Abstract	1
2	Nomenclature	1
3	Introduction	1
4	Experimental Apparatus and Procedures	3
4.1	Experimental Apparatus	3
4.2	Operating Technique	3
5	Results	3
5.1	Operating Conditions	3
5.2	Measurements	3
5.3	Quantities of Interest	3
6	Analysis and Discussion	3
6.1	Data Conversion	3
6.2	Interpretation	3
7	Summary and Conclusions	3
8	References	3
9	Appendices	3
9.1	Data	3
9.2	Code	4

Figures

1	Superposition of uniform flow and a doublet to model nonlifting potential flow over a circular cylinder.	1
---	--	---

Tables

1 Abstract

2 Nomenclature

3 Introduction

For the inviscid-flow model used in this lab, we assume a potential-flow field that is:

- **Incompressible (continuity):** $\nabla \cdot \vec{V} = 0$.
- **Irrotational:** $\nabla \times \vec{V} = 0$.

If $\nabla \times \vec{V} = 0$, then a *velocity potential* ϕ exists such that $\vec{V} = \nabla \phi$. Substituting into continuity gives Laplace's equation:

$$\nabla^2 \phi = 0 \quad \left(\text{i.e., } \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \right).$$

In order to model flow over a 2D cylinder, we use an approach presented in Anderson's Fundamentals of Aerodynamics, which uses a superposition of a uniform flow and a doublet flow, as presented in the figure below:

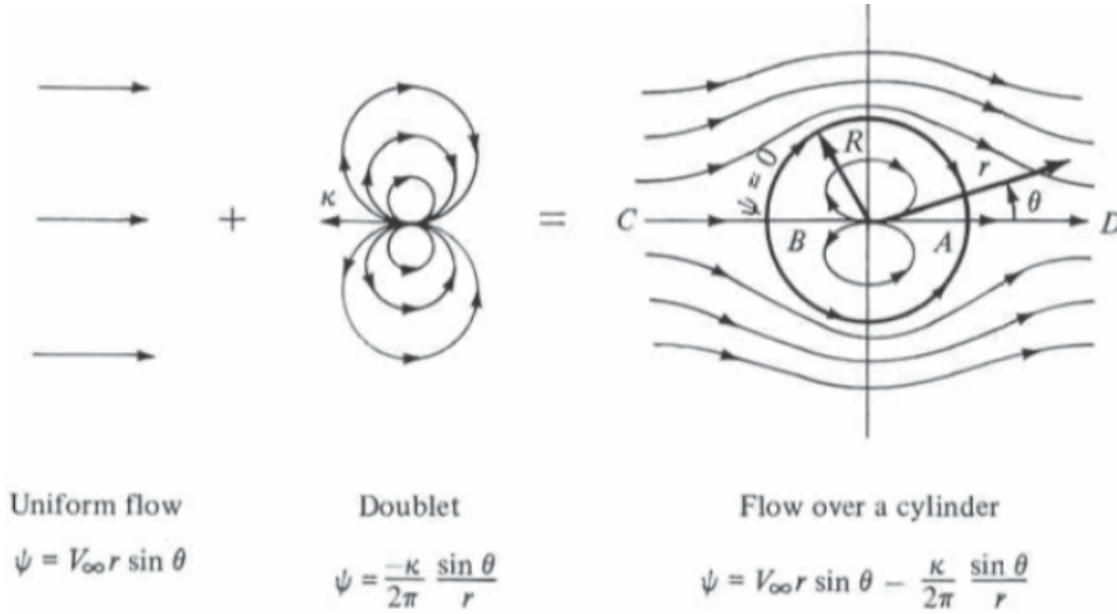


Figure 3.26 Superposition of a uniform flow and a doublet; nonlifting flow over a circular cylinder.

Figure 1: Superposition of uniform flow and a doublet to model nonlifting potential flow over a circular cylinder.

In polar coordinates (r, θ) , the stream function and velocity potential for both the uniform flow and the doublet are given by:

$$\phi_U = V_{\infty} r \cos(\theta) \quad (\text{velocity potential: uniform flow}) \quad (1)$$

$$\psi_U = V_{\infty} r \sin(\theta) \quad (\text{stream function: uniform flow}) \quad (2)$$

$$\phi_D = \frac{\kappa}{2\pi} \frac{\cos(\theta)}{r} \quad (\text{velocity potential: doublet}) \quad (3)$$

$$\psi_D = -\frac{\kappa}{2\pi} \frac{\sin(\theta)}{r} \quad (\text{stream function: doublet}) \quad (4)$$

where V_∞ is the freestream speed and κ is the doublet strength. Superposing the uniform flow and doublet stream functions gives

$$\begin{aligned} \psi &= \psi_U + \psi_D \\ &= V_\infty r \sin(\theta) - \frac{\kappa}{2\pi} \frac{\sin(\theta)}{r} \\ &= V_\infty r \sin(\theta) \left(1 - \frac{R^2}{r^2} \right), \end{aligned} \quad (5)$$

where the cylinder radius is defined by

$$R^2 \equiv \frac{\kappa}{2\pi V_\infty}.$$

The corresponding velocity field in polar coordinates follows from the stream function relations $V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta}$ and $V_\theta = -\frac{\partial \psi}{\partial r}$:

$$V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = \frac{1}{r} (V_\infty r \cos(\theta)) \left(1 - \frac{R^2}{r^2} \right) = \left(1 - \frac{R^2}{r^2} \right) V_\infty \cos(\theta), \quad (6)$$

$$V_\theta = -\frac{\partial \psi}{\partial r} = -\left(1 + \frac{R^2}{r^2} \right) V_\infty \sin(\theta). \quad (7)$$

At the surface of the cylinder, $r = R$, the velocity field is:

$$V_r = \left(1 - \frac{R^2}{R^2} \right) V_\infty \cos(\theta) = 0, \quad (8)$$

$$V_\theta = -\left(1 + \frac{R^2}{R^2} \right) V_\infty \sin(\theta) = -2V_\infty \sin(\theta). \quad (9)$$

This aligns with the assumption that the velocity normal to the stationary surface must be zero, and due to polar coordinates, the velocity normal to the surface is given by V_r . Since V_θ is perpendicular to the surface, due to construction of the coordinate system, it's equivalent to a tangential velocity, and since normal velocity is zero, $V = V_\theta = V_\infty \sin(\theta)$.

The pressure coefficient is given by:

$$C_p = 1 - \frac{V^2}{V_\infty^2}$$

$$V = -2V_\infty \sin(\theta) \quad (10)$$

$$\frac{V^2}{V_\infty^2} = 4 \sin^2 \theta \quad (11)$$

$$\boxed{C_p = 1 - \frac{V^2}{V_\infty^2} = 1 - 4 \sin^2 \theta} \quad (12)$$

4 Experimental Apparatus and Procedures

4.1 Experimental Apparatus

4.2 Operating Technique

5 Results

5.1 Operating Conditions

5.2 Measurements

5.3 Quantities of Interest

6 Analysis and Discussion

6.1 Data Conversion

6.2 Interpretation

7 Summary and Conclusions

8 References

References

- [1] W. Labov, *Sociolinguistic Patterns*. Philadelphia: University of Pennsylvania Press, 1972.
- [2] N. Chomsky, *Syntactic Structures*. The Hague: Mouton, 1957.

9 Appendices

9.1 Data

`./data/pressure_vs_alpha.csv`

```
1 | alpha_deg,P_inf,P_0,P
2 | 0,15.4,23.0,23.0
3 | 5,15.4,23.1,22.7
4 | 10,15.4,23.1,22.3
5 | 15,15.4,23.2,21.3
6 | 20,15.3,23.1,19.9
7 | 25,15.4,23.1,18.6
8 | 30,15.4,23.2,16.9
9 | 35,15.4,23.1,15.0
10 | 40,15.4,23.1,13.1
11 | 45,15.4,23.1,11.2
12 | 50,15.3,23.1,9.4
13 | 55,15.3,23.1,8.4
14 | 60,15.4,23.1,7.4
15 | 65,15.4,23.1,7.0
```

```
16 70,15.4,23.2,6.9
17 75,15.4,23.2,7.6
18 80,15.4,23.2,7.8
19 85,15.4,23.1,8.0
20 90,15.4,23.1,8.1
21 95,15.4,23.1,8.3
22 100,15.4,23.2,8.3
23 105,15.4,23.1,8.2
24 110,15.4,23.1,8.1
25 115,15.4,23.1,8.1
26 120,15.4,23.1,8.1
27 125,15.4,23.1,7.9
28 130,15.4,23.1,7.8
29 135,15.4,23.1,7.8
30 140,15.4,23.1,7.7
31 145,15.4,23.1,7.6
32 150,15.4,23.1,7.7
33 155,15.4,23.1,7.8
34 160,15.4,23.0,8.0
35 165,15.4,23.1,8.0
36 170,15.4,23.1,8.2
37 175,15.4,23.1,8.2
38 180,15.4,23.1,8.2
```

9.2 Code